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Paleomagnetic and structural evidence for localized tectonic rotation associated with fault drag in the northeastern Mojave Desert: Implications for the late Cenozoic tectonic evolution of the Eastern California shear zone

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Abstract

Paleomagnetic data, coupled with detailed geological analysis of the southeastern Goldstone Lake region, indicate that lower Miocene volcanic and epiclastic rocks of the Pink Canyon area have been folded, faulted and tectonically rotated $\sim 28.4 \pm 9.0°$ clockwise about a vertical axis; identical rocks lying to the west across the Goldstone Lake Fault (east branch) have been rotated $9.6 \pm 7.4°$. The Coyote Canyon Fault is locally folded about a vertical axis $\sim 25°$ in a clockwise sense in the Pink Canyon area. Timing relationships indicate that rotation is post-early Miocene; regional relationships imply that deformation is late Miocene to Holocene in age. These relationships imply that tectonic rotation is local rather than regional in extent as proposed by some tectonic models. The results of this study are generally consistent with the Dokka and Travis [1] model and the subsequently revised Dokka model [2] of strain partitioning in the northeastern Mojave Desert block.

1. Introduction

The Goldstone Lake region is located in the northeastern part of the Mojave Desert block within the Eastern California shear zone (ECSZ) of Dokka and Travis [1] (Fig. 1). The ECSZ is important because it has and continues to accommodate between 9 and 23% of the total right shear between the Pacific and North American plates since its inception at ~ 10 Ma [3,4]. The tectonic history of the Goldstone Lake region has only been poorly understood until recently due to a paucity of geological and geophysical information. In addition, access to this area has been limited as much of it lies within the boundaries of the Army National Training Center and the Jet Propulsion Laboratory's Goldstone Deep Space Communication Complex. Due to lack of data, early models of the late Cenozoic tectonic history of this region were simplistic, relying on little geological information. More recent models incorporated more geological and geophysical data but were still somewhat speculative due to data limitations. One of the key differences between early and more recent models is the prediction of homogeneous regional or localized tectonic rotations, respectively. This paper presents the results of paleomagnetic studies coupled with detailed geological analysis of the southeastern Goldstone Lake region and provides evidence that late

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Cenozoic vertical-axis rotations are local and not regional in nature. Our work suggests that the local rotation is related to a presently active strike-slip fault and that rotation may be occurring today.

2. Regional tectonic models

Several regional tectonic models have been proposed to explain the late Cenozoic strain in the Mojave Desert region. Garfunkel [5], based on generalized fault orientations and displacements (Fig. 2a), proposed that uniform stresses acting over the entire Mojave Desert region resulted in a homogeneous E–W trending simple sinistral shear system linked to extension in the Basin and Range Province. He reasoned that fault blocks originally oriented sub-perpendicular (ca. N–S) to the shear couple were rotated counterclockwise by as much as 30°. Areas where the faults were predominantly E–W trending, as was thought to be the case in the northeast Mojave Desert region, would not have been subject to counterclockwise rotation as left slip alone would have accommodated the strain. However, in the case of the northeastern Mojave desert, Garfunkel proposed that clockwise rotation had resulted from a local deviation in homogeneous shear expressed in the bending of the Garlock fault. Later models by others [6–8] also invoked 'domino-style' tectonics (Fig. 2b). These models were based on paleomagnetic data from limited areas. However, no paleomagnetic studies were conducted in the northeastern Mojave Desert region. These models predicted between ~ 40° [8] and ~ 80° [6,7] of clockwise rotation for the northeastern Mojave Desert region and small amounts of counterclockwise rotation (~ 3–12°) for the greater Mojave Desert region; all deformation was thought to be linked to dextral shear caused by Pacific–North American plate interactions.

The above models predict some degree of regional clockwise rotation in the northeastern Mojave Desert region. In contrast, the model of Dokka and Travis [1] and the subsequent revision [2] (Fig. 2c) predict no regional rotation in the northeastern Mojave Desert; however, the occurrence of local clockwise rotations of < 40° in association with strike-slip faults was thought possible. Dokka and Travis [1] proposed that

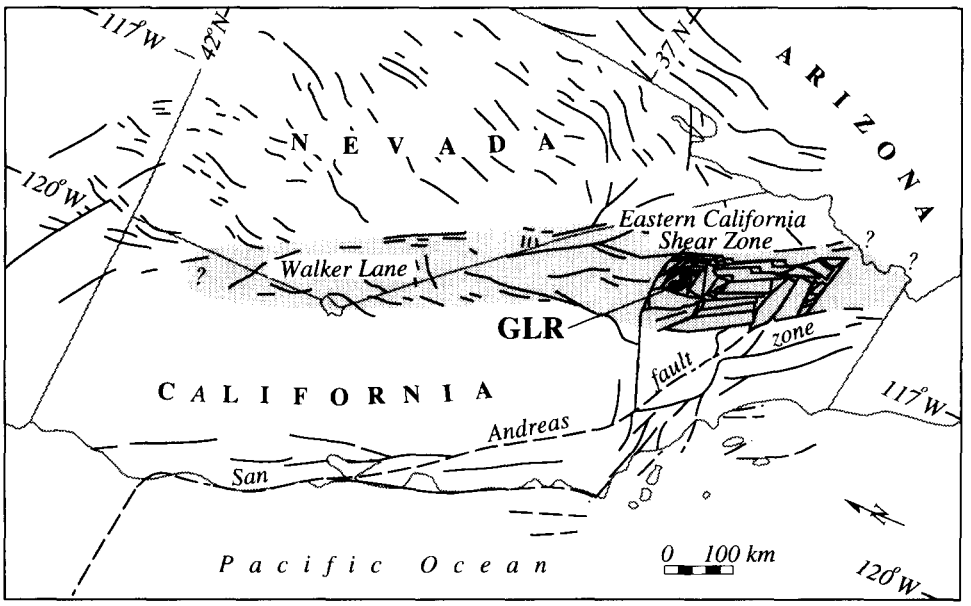


Fig. 1. Schematic map highlighting the Eastern Califronia shear zone and Goldstone Lake region (*GLR*).

~ 57 km of right shear has been distributed in a structurally complex terrane between the Blackwater Fault and the Mesquite Valley disturbed zone. This deformation accommodated the oroclinal bending of the eastern Garlock fault and resulted in substantial heterogeneous strain across the region.

3. Geological setting

The general geology and stratigraphy of the study area are illustrated in Fig. 3. The rocks include Mesozoic intrusions and metamorphic basement [9], Tertiary volcanic rocks and epiclastic strata, and basin-filling Quaternary fanglomer-

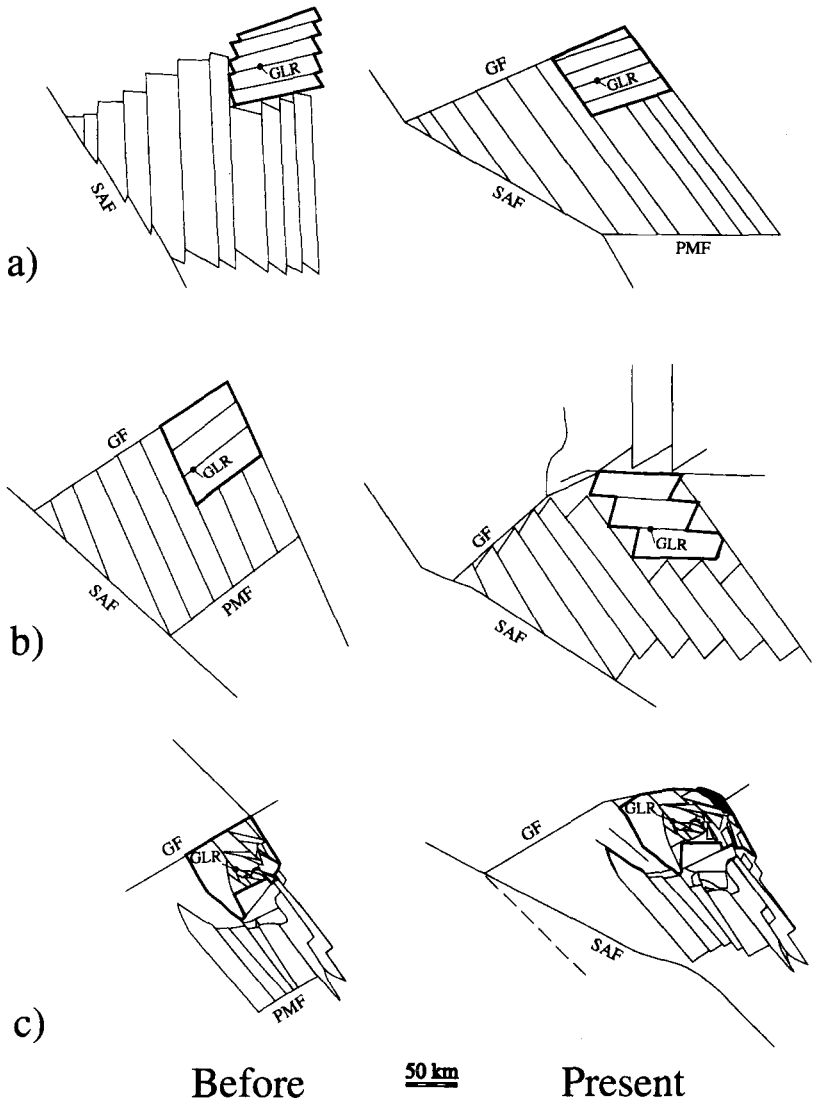


Fig. 2. Models for the late Cenozoic tectonic evolution of the Mojave Desert region. Modified from (a) Garfunkel [5], (b) Carter et al. [8] and (c) Dokka [3]. *GF* = Garlock fault; *GLR* = Goldstone Lake region; *PMF* = Pinto Mountain Fault; *SAF* = San Andreas Fault. Area within heavy outline is the northeastern Mojave Desert region. Scale is approximate.

ate, lacustrine and alluvial sediments. $^{40}Ar/^{39}Ar$ age spectrum dating of Tertiary strata (Fig. 3) conducted by A.K. Baksi [pers. commun., 1994] following the methods outlined in Baksi [10] yielded a date of 18.4 ± 0.2 Ma (whole-rock, stepwise) on a formation capping basalt (Fig. 4), and a date of 22.3 ± 1.5 Ma (hornblende, total fusion) from rhyolite air-fall pumice clasts.

Two major faults occur in the study area, the NW-striking right-slip Goldstone Lake Fault Zone and the ca. E–W-striking left-slip Coyote Canyon Fault (Fig. 3); both faults cut unconsolidated Quaternary alluvium and have a component of dip slip. Where exposed, most fault segments are steeply dipping. The sense of slip on both of these faults is inferred from kinematic indicators such as slickenside striations, oriented mineral growths, and lithological offsets, as well as from the tectonic setting. These faults intersect immediately southwest of the Pink Canyon area (Fig. 3). Rocks northeast of this intersection are folded into a broad, gently NNE-plunging antiform, the form of which is reflected in the physiography of the area. The strike of the Goldstone Lake fault (east branch) is apparently unaffected at its junction with the Coyote Canyon Fault. In contrast, the Coyote Canyon Fault is apparently folded about a sub-vertical axis, curving toward the north, near the Goldstone Lake Fault (east branch); the change in strike is ~ 25°.

4. Paleomagnetic data

Oriented samples for paleomagnetic analysis were collected in the field using a portable drill

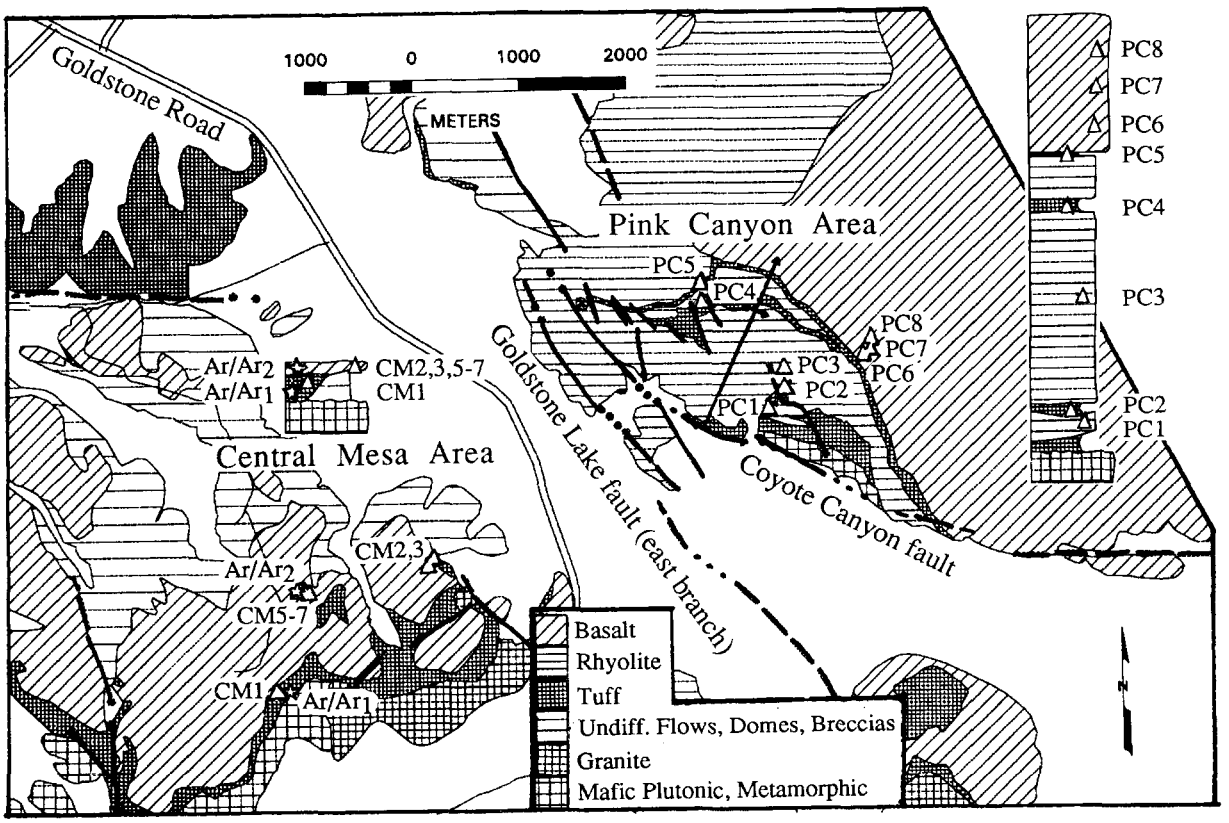


Fig. 3. Geological map and columnar sections of the southeast Goldstone Lake region showing paleomagnetic and $^{40}Ar/^{39}Ar$ age dating sample sites. Note fold axis in Pink Canyon area. UTM coordinates for all paleomagnetic sample sites are given in Table 1.

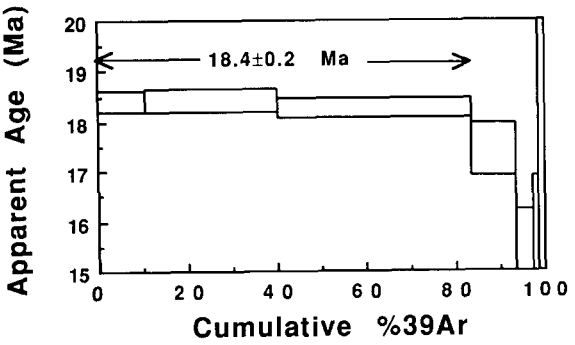


Fig. 4. $^{40}Ar/^{39}Ar$ age spectrum for whole-rock basalt from the Central Mesa area. Errors in step ages shown at one standard deviation, internal precision. Plateau age reflects an uncertainty of ±1.0% in the age of the monitor.

fitted with a 1″ coring bit. In addition, oriented blocks sampled by hand and cored in the laboratory with a drill press were also used. All orientations were made with a magnetic compass and clinometer. Site localities were chosen where the structure and stratigraphy were well known from previous detailed geological mapping. Samples were collected from fifteen sites, seven from the mesas (hereafter referred to as the Central Mesa area) located west of the Goldstone Lake Fault Zone (east branch) and eight from the Pink Canyon area located east of the Goldstone Lake Fault Zone (east branch) (Fig. 3). Samples were taken from basalt flows, a rhyolite flow, a rhyolite dike, a basaltic pyroclastic fall deposit and volcaniclastic rocks. Table 1 gives the paleomagnetic data and pertinent statistics.

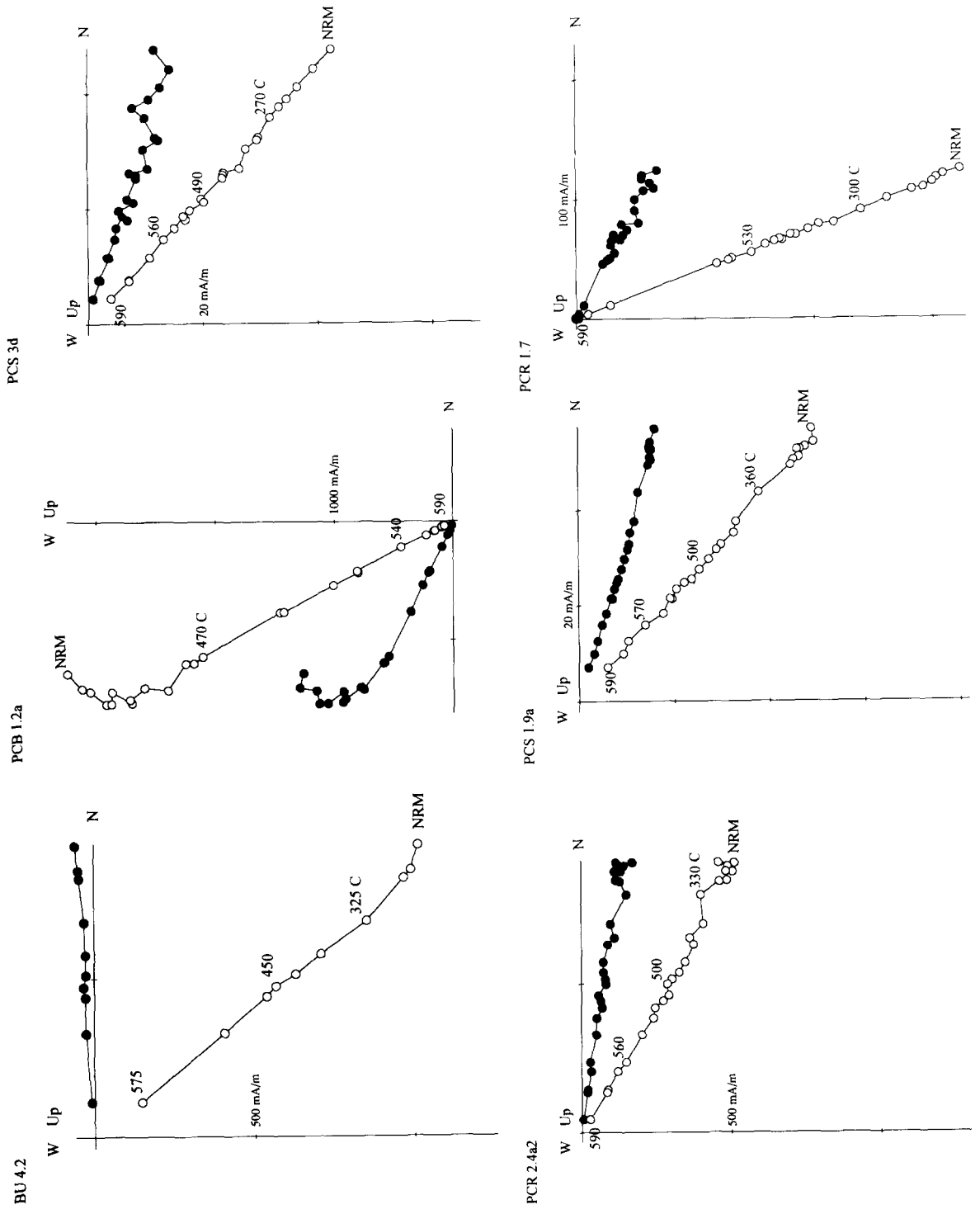
Natural remanent magnetization (NRM) directions were measured at Louisiana State University for all specimens using a cryogenic magnetometer. Alternating field and thermal demagnetization analyses on a few test specimens showed similar results. The rest of the specimens were analyzed using standard partial thermal demagnetization techniques. All specimens were demagnetized in nine or more steps. Orthogonal vector end-point diagrams were constructed for each specimen to evaluate the magnetic components present and to define the characteristic remanent magnetization (ChRM). Many samples

Table 1
Site mean paleomagnetic directions of the characteristic magnetization

Map Index	Site Name	Rock Type	UTM (easting-northing)	N/N_o	Bedding (S/D)	In Situ (D/I) deg	Fold Corrrected (D/I) deg	Plunge Corrected# (D/I) deg	a_{95}	k
				Central Mesa						
CM1*	EV1	conglomerate	513710-3907150	6/6	180/16	N/A	N/A	N/A	120.8	1.3
CM2	BU1-6	basalt	515300-3908400	18/18	218/03	009.1/61.8	004.5/60.2	N/A	2.2	240.9
CM3	BU7-8	basalt	515200-3908400	6/6	218/03	356.1/58.8	352.7/56.7	N/A	4.6	209.7
CM4	CC1-2	basalt	514000-3908000	6/6	305/05	005.6/63.3	009.8/58.9	N/A	3.5	360.7
CM5	CC4	basalt	514000-3908000	3/3	305/05	354.9/58.6	359.6/54.6	N/A	4.3	812.1
CM6	CC5	basalt	514000-3908000	3/3	305/05	348.1/55.8	352.9/52.2	N/A	4.0	933.7
CM7	CC6	basalt	514000-3908000	3/3	305/05	358.8/51.9	002.2/47.8	N/A	4.4	800.6
				Pink Canyon						
PC1	PCR1	rhyolite	518410-3909780	14/15	288/15	029.6/67.5	020.1/53.4	25.5/52.7	3.4	139.4
PC2	PCS1	sandstone	518560-3909970	19/19	289/13	016.1/50.1	016.7/37.1	16.8/37.1	2.4	191.7
PC3	PCR2	rhyolite	518560-3910150	21/21	219/22	024.2/42.2	008.2/33.5	11.0/33.5	4.4	53.6
PC4	PCS3	sandstone	517830-3910830	5/5	257/23	036.4/61.3	017.0/43.3	18.9/43.3	4.3	315.0
PC5	PCS2	air-fall tuff	517780-3910950	8/10	237/33	031.7/58.0	003.3/36.0	07.0/36.0	6.5	72.5
PC6	PCB3	basalt	519300-3910270	8/8	329/24	358.9/70.0	030.8/52.7	29.0/52.7	5.0	125.9
PC7	PCB2	basalt	519370-3910320	8/8	329/24	010.1/65.9	032.5/46.4	30.7/46.4	2.9	366.2
PC8	PCB1	basalt	519370-3910430	4/4	329/24	332.0/70.5	019.1/58.7	17.4/58.7	6.1	228.6

Means	n	In Situ	a95(In Situ)	k(In Situ)	Corrected	a_{95}(corrected)	k(corrected)	R^	F^
Central Mesa	6	358.3/58.5	4.7	203.9	00.0/55.2	4.9	187.5	9.6 ± 7.4	-1.7 ± 4.6
Pink Canyon	8	018.1/61.9	9.4	35.9	18.8/45.3	7.5	55.1	28.4 ± 9.0	8.2 ± 6.4

#Corrected for both folding and plunge of fold axis ($D/I = 24°/15°$); *conglomerate test site, $R = 2.11$; R and F calculated following Demarest [17]; reference direction calculated from the Miocene reference pole of Diehl et al. [15] (pole lat. = 82.0N, long = 146.9E, $A_{95} = 2.9$).


PCS 3d
N
W
Up
NRM
270 C
490
560
590
20 mA/m
PCR 1.7
100 mA/m
300 C
530
PCB 1.2a
1000 mA/m
NRM
470 C
540
590
PCS 1.9a
20 mA/m
360 C
500
570
BU 4.2
500 mA/m
325 C
450
575
PCR 2.4a2
330 C
500
560

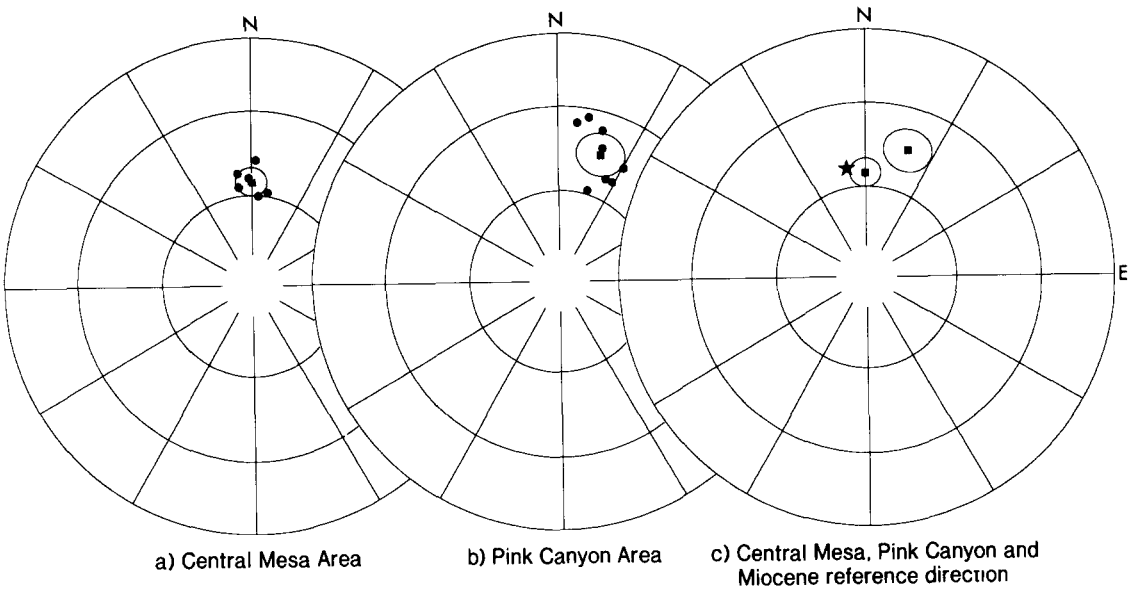


Fig. 6. Equal-area projections showing structurally corrected paleomagnetic directions from the Central Mesa and Pink Canyon sites (●); means (■) shown with 95% confidence limits (circles) compared to the expected direction for stable Miocene North America (★) (recalculated from Diehl et al. [15]).

showed one or more low unblocking temperature component(s). A higher unblocking temperature component which showed nearly univectorial decay toward the origin of the diagrams was selected as representative of the ChRM direction. Representative orthogonal vector end-point diagrams are shown in Fig. 5.

With the exception of one site (PCB1) (map index = PC8), all site directions exhibit normal polarity. Most samples were strongly demagnetized by ~580°C, suggesting that the ChRM component primarily resides in magnetite; however, almost all specimens had a small high unblocking temperature component that remained after heating to this level. This remaining component is generally in the ChRM direction, as can be seen from the orthogonal vector end-point diagrams (Fig. 5), and indicates the presence of a minor early-acquired magnetization residing in hematite. The ChRM is taken to be the primary magnetization. In the case of the igneous rocks, the ChRM is most likely a thermal remanent magnetization (TRM) acquired when the rocks cooled through the unblocking temperature of the magnetic mineral grains. For volcaniclastic rocks, the ChRM likely represents detrital remanent magnetization (DRM) acquired when the sediments were deposited.

Application of the correlation fold test [11] on all sites from the Pink Canyon area showed a significant correlation for in-situ directions and no correlation for structurally corrected directions, indicating a passing of the fold test at the 99% confidence level. The optimal test statistic was found at 56% unfolding; we consider this to be an artifact of the primary dips of volcaniclastic and rhyolite flow strata rather than evidence for syn-folding remagnetization. The inclinations of site mean directions are somewhat steeper than the expected inclination for the site before struc-

Fig. 5. Orthogonal end-point diagrams showing examples of basalts (BU 4.2, PCB 1.2a), volcaniclastics (PCS 3d, PCS 1.9a) and rhyolites (PCR 2.4a2, PCR 1.7). ● = vector end points projected onto the horizontal plane; ○ = vector end points projected onto the vertical plane.

tural correction, and shallower after structural correction. Again, this may be the result of a small component of primary dip. However, the strike of the bedding is such that the mean declination for the Pink Canyon area is not significantly affected by the structural correction (before correction $D = 18.1°$; after correction $D = 18.8°$).

At site EV1 (map index = CM1) six individual pumice clasts were sampled from an outcrop of massive 'tuff breccia' and were analyzed using partial thermal demagnetization techniques. The Watson [12] test for randomness of directions was applied to the resulting specimen mean data. The EV1 samples passed the test for randomness at the 95% confidence level. This suggests that these pumice clasts were deposited as a matrix-supported volcaniclastic conglomerate (possibly of laharic origin) and that this unit has not been strongly remagnetized since deposition. This conclusion is corroborated by the small-scale, basin-filling, onlap geometry of the deposit. The interpretation that EV1 has not been remagnetized implies that none of the sites has been substantially remagnetized by a regional chemical/thermal event, and that the ChRM directions of all sites have likely remained stable since deposition.

Directional independence tests were performed on the basalt flows from the Central Mesa area to define 'cooling units' as described by Calderone et al. [13]. The mean direction for each site/flow was compared to the stratigraphically adjacent site/flow using the statistical test derived by McFadden and Lowes [14] for two populations having unequal precision parameters. Six distinct 'cooling units' were identified at the 99% confidence level. Thus, fourteen individual sites were sampled, but were then reduced to six distinct 'cooling units', referred to as BU1-6, BU7-8, CC1-2, CC4, CC5 and CC6 (see Table 1 for map indices). This method should result in a mean direction (and associated statistical parameters) that more closely approximates an independent sampling of the paleofield [13].

Structurally corrected paleomagnetic directions for all sites, except EV1, are compared to the expected direction for stable Miocene North America (recalculated from Diehl et al. [15]) in Fig. 6. Mean directions were determined independently for the Pink Canyon sites and the Central Mesa sites. The Pink Canyon and Central Mesa mean directions do not overlap at the 99% confidence level using the statistical discrimination test described by McFadden and Lowes [14]. After structural correction, the means are $D/I = 18.8°/45.3°$ ($\alpha_{95} = 7.5°$) and $0.0°/55.2°$ ($\alpha_{95} = 4.9°$) for the Pink Canyon and Central Mesa areas, respectively, using the statistical methods of Fisher [16]. A statistically significant clockwise declination anomaly of $\sim 28°$ exists between the Miocene reference direction and the site-mean direction for the Pink Canyon area, but the declination anomaly for the Central Mesa area is marginally significant at the 95% confidence level. The rotation and flattening statistics of Demarest [17] yield a rotation R of $28.4 \pm 9.0°$ and a flattening F of $8.2 \pm 6.4°$ for the Pink Canyon area and $R = 9.6 \pm 7.4°$ and $F = -1.7 \pm 4.6°$ for the Central Mesa area (see Table 1). The somewhat anomalous inclination in the Pink Canyon mean direction is interpreted to be due to unaccounted for primary dips of volcaniclastic rocks and rhyolite flows (see above discussion on structural corrections). As there are no flow or 'cooling units' that can be uniquely correlated across the fault, the rotation and flattening statistics are given with respect to the Miocene reference field rather than between the two areas. The results indicate that significant differential rotation has occurred between the Pink Canyon and Central Mesa areas.

The angular dispersion of the site-mean directions, the presence of a reversed-polarity site, and the age range of the volcanic sequence suggest that the secular variation of the geomagnetic field has been at least partially averaged in this study. Based on this interpretation we have applied the geocentric axial dipole field hypothesis to our results in the following discussion.

5. Discussion and conclusion

Paleomagnetic data from the Goldstone Lake region have important implications for the late Cenozoic tectonic evolution of the Eastern Cali-

fornia shear zone. Data from both sides of the Goldstone Lake Fault (east branch) suggest differential rotation across the fault. Mean paleomagnetic directions from rocks west of the Goldstone Lake Fault (east branch) in the Central Mesa area are marginally distinguishable from the Miocene reference direction, whereas mean directions from rocks in the Pink Canyon area to the east of the fault are rotated ~ 28° clockwise. The magnitude and sense of vertical-axis rotation found in the Pink Canyon area (~ 28° clockwise) is similar to the measured angle of deflection (~ 25° clockwise) of the westernmost segment of the Coyote Canyon Fault near its junction with the Goldstone Lake Fault (east branch) (Fig. 3). We propose that both the clockwise tectonic rotation of the rocks in the Pink Canyon area and the northward folding of the western segment of the Coyote Canyon Fault are the result of localized fault drag along the right-slip Goldstone Lake Fault (east branch). These relationships are consistent with the models of Dokka and Travis [1] and Dokka [2] which state that post-18 Ma vertical-axis rotations in the northeastern Mojave Desert are local in extent and associated with strike-slip faulting. The data do not support the models of Garfunkel [5], Luyendyk et al. [6,7] and Carter et al. [8] that predict regionally homogeneous vertical-axis rotations in the northeastern Mojave Desert. The present-day seismic activity of the Goldstone Lake Fault Zone suggests that local vertical-axis rotation may be currently active.

Acknowledgments

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